

Fundamental Theorem of Algebra (2.6)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $-4i$

_____ How many complex zeros (with multiplicity) does a degree-5 polynomial have?

ANSWER: $-\sqrt{5}$ # _____ How many complex zeros does $f(x) = x^4 + 1$ have?ANSWER: $2 - 3i$ # _____ If $1 - i$ is a zero (real coefficients), name another zero.

ANSWER: 0

_____ Write the factored form of a cubic with a double zero at $x = 1$ and a zero at $x = -2$.

ANSWER: 5 # _____ If $2 + 3i$ is a zero of a polynomial with real coefficients, what other zero must it have?	ANSWER: $x^2 + 9$ # _____ How many real zeros does $f(x) = x^2 + 4$ have?
ANSWER: $x(x - 2)(x + 2)$ # _____ If $\sqrt{5}$ is a zero of a polynomial with rational coefficients, name another zero.	ANSWER: $(x - 1)^2(x + 2)$ # _____ What is the conjugate of the zero $4i$?
ANSWER: $1 + i$ # _____ Write the factored form of a cubic with zeros 0, 2, and -2.	ANSWER: 4 # _____ Write a quadratic factor for a polynomial with zeros $3i$ and $-3i$.